

Yaohui Shu

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SUMMARY

Applied mathematician and statistician working in stochastic modeling, numerical PDE, and statistical inference. Derives closed-form structure from stochastic differential equations, turns it into stable solvers verified against analytical identities to machine precision, and estimates causal quantities from noisy, nonstationary, heavy-tailed data. M.S. Finance, CQF.

EDUCATION

Stony Brook University — Ph.D., Applied Mathematics & Statistics Expected May 2027 GPA: 3.78 / 4.0
University of Rochester — M.S., Finance Aug 2021 – Aug 2022
Derivatives pricing, financial econometrics, risk management, stochastic calculus for finance. GPA: 3.81 / 4.0
CQF Institute — Certificate in Quantitative Finance (CQF)

SKILLS

Programming Python (NumPy, SciPy, Pandas, TensorFlow), MATLAB, C++, R; SLURM/HPC, vectorized and parallel computing
Numerical Methods Finite-volume and finite-difference PDE solvers, upwind and Chang-Cooper schemes, sparse and low-rank linear algebra (Woodbury), Monte Carlo, calibration by root-finding, nondimensionalization and stiffness control
Stochastic Modeling Stochastic calculus, SDE / Fokker-Planck / Langevin dynamics, Ornstein-Uhlenbeck processes, absorbing boundaries and first-passage, martingales and Wald identities, nonstationary and regime-switching models
Inference & ML Causal discovery, k-nearest-neighbor conditional independence testing, doubly robust estimation (DML, AIPW, TMLE), transfer entropy, LSTM, XGBoost, GNN, PCA
Finance Derivatives pricing, financial econometrics, equity signal research, backtesting

RESEARCH & EXPERIENCE

Stony Brook University *Stony Brook, NY*
Dissertation Research: Stochastic Modeling, Numerical PDE & Causal Inference 2024 – Present

- Solved the stationary two-dimensional Fokker-Planck equation for a degenerate hypoelliptic transport-diffusion system: conservative finite volume, second-order upwind transport, Chang-Cooper diffusion preserving detailed balance, and a half-range absorbing boundary closed by a maximum-entropy re-injection kernel.
- Recast the boundary recycling term as a low-rank update solved by the Woodbury identity, factorizing only the local operator and matching the direct coupled solve to machine precision; outer bisection enforces a nonlinear self-consistency constraint.
- Derived closed-form structural identities (Gaussian marginal, current profile, a generalized Wald identity, an existence bound) and used them as solver certificates for conservation and convergence order.
- Reduced the physical parameter space to two dimensionless groups by nondimensionalization, curing numerical stiffness and collapsing the full parameter sweep onto master curves.
- Designed a causal discovery framework for feedback systems with unmeasured drivers: partial ancestral graphs under sigma-separation replacing acyclicity-constrained methods (NOTEARS, DYNOTEARS), with distribution-free k-nearest-neighbor independence tests.

Teaching Assistant, AMS 595 Fundamentals of Computing (MATLAB, Python, C++) Aug 2026 – Present

Brookhaven National Laboratory *Upton, NY*
Research Assistant: Stochastic Processes, Statistical Inference & Information Theory Jan 2024 – Jan 2026

- Modeled coupled Ornstein-Uhlenbeck systems from white-noise to long-memory regimes with convergence and stationarity proofs, and derived exact Fokker-Planck eigenspectra and closed-form stationary distributions.
- Derived closed-form, time-resolved information-flow estimators and an ensemble-based pointwise estimator recovering transient structure that time-averaged methods miss; validated against 10^5 -path Monte Carlo ensembles on HPC/SLURM pipelines.

CQF Institute — Quantitative Finance Capstone *Remote*
Equity Return Prediction: LSTM, Feature Engineering & Signal Research 2023

- Built an end-to-end pipeline for equity-return direction prediction from 156+ engineered momentum, volatility, and technical features, reduced to 10–15 predictors via XGBoost importance, self-organizing maps, and PCA; leakage-free splits, systematic hyperparameter search, and out-of-sample monitoring.

PUBLICATIONS

- Shu, Y.**, Liu, Y., Zhang, T., McGraw, R., Li, X. (2026). Stochastic Condensation with Information Flow for Causality Analysis. *Geophysical Research Letters*, 53, e2026GL123056. DOI: 10.1029/2026GL123056
- Shu, Y.** Entropy-Based Causal Discovery for Complex Dynamical Systems (working paper, 2026); other three working papers in preparation.